Bench Instruments Guide

This guide covers the bench instruments at each lab station: how to set their limits and outputs, and how to take reliable measurements with the oscilloscope, the Fluke 39 power meter and the Acuvim-L panel meter. Each lab handout gives the values to use (voltages, current limits, protection levels and starting oscilloscope scales); this guide gives the front-panel procedures. Read it before your first hardware lab and keep it open at the bench.

Set every limit before the instrument is wired into the circuit, with its output off. Front-panel keys are shown in [brackets].

Instrument	Used for	Section
HP/Keysight E3631A triple-output supply	+15 V auxiliary power for the inverter box and interface board. Never part of the power circuit.	1
BK Precision 1666 (1–40 V, 0–5 A)	DC input supply	2
ITECH IT-M3632 (60 V, 30 A, 800 W, bidirectional)	Electronic load or DC source	3
Keysight DSOX2004A oscilloscope, N2791A differential voltage probe, 1146B current probe	Waveform capture and measurements	4
Fluke 39 power harmonics meter with i400s current clamp	Harmonics, THD and power at the fundamental	5
Accuenergy Acuvim-L panel power meter with 30:5 A CTs	Three-phase rms values, power and harmonics at the station transformer	6

1. E3631A — +15 V auxiliary supply

- Use the +25V output: its + terminal and COM go to the +15 V and GND jacks on the interface board.
- Press [+25V], then [Display Limit] (the Lmt annunciator blinks). Press [Vol/Cur] to select current and turn the knob to 1.000 A; the arrow keys choose which digit the knob changes. Press [Vol/Cur] again and set 15.00 V. Press [Display Limit] to return to the meter display.
- [Output On/Off] switches all three outputs of the E3631A together.

2. BK Precision 1666 — DC supply

- **The BK 1666 has no output on/off key: its output is live whenever the power switch is on.** When a handout says to turn the BK off, use the power switch.
- Voltage: set with the Voltage Coarse and Fine knobs while reading the voltmeter. Start with both knobs fully counter-clockwise, and return them fully counter-clockwise before switching the supply off at shutdown.
- Current limit: turn both Current knobs (Coarse and Fine) fully clockwise. This gives the supply's maximum limit of 5 A, which is the setting used unless a handout specifies a lower limit.

3. ITECH IT-M3632 — electronic load and DC source

- Values are entered after pressing a setting key, with the knob or numeric keys, and confirmed with [Enter]. [On/Off] turns the input (load mode) or output (source mode) on and off; the key is lit when on. Check the mode shown on the display before turning it on.
- Load mode, constant resistance (CR): press [Load]. Press [Shift] + [V-set] (Config), select Mode and press [Enter], select CR and press [Enter], then [Esc]. Press [R-set] and enter the resistance.
- Source mode: press [Source]. Press [V-set] and enter the voltage, then [I-set] and enter the current limit.
- Over-voltage protection (OVP): press [Shift] + [R-set] (Protect), select OVP and press [Enter], select On and press [Enter], set Level and press [Enter], leave Delay at its default and press [Enter], then [Esc].
- **OVP only watches the ITECH's own terminals.** It does not protect a converter output elsewhere in the circuit, such as a boost converter's DC link.

4. Oscilloscope — Keysight DSOX2004A

Do not use [Auto Scale]. On a switching converter it rescales every channel, moves the trigger, and switches off any channel with no signal yet (for example the current channel before the load is on), so the traces you set up disappear or become unreadable. If you press it by accident, press the Undo AutoScale softkey before any other key. Set the scope up by hand as follows.

Probes

- N2791A differential voltage probe: select the attenuation on the probe (10:1 unless the handout says otherwise) and check that its input range covers the voltage you will measure.
- 1146B current probe: switch it on, select the 100 mV/A range (or 10 mV/A for larger currents), and zero it with the zero adjust before clamping it around the

wire, with the arrow on the jaw pointing in the direction of positive current. Switch it off after the lab to save its battery.

- Tell the scope what is connected: channel key → [Probe]. Check this on every channel each session: the scope does not detect these probes and keeps whatever setting the previous user left. Voltage channel: Ratio 10:1 to match the probe. Current channel: Units = Amps and 0.1 V/A for the 100 mV/A range (0.01 V/A for 10 mV/A). The scope then reads directly in volts and amps.

Scales, coupling and trigger

- The screen has 10 horizontal and 8 vertical divisions. Choose a time/div that shows what you are measuring (for example two to three switching periods for ripple), and a V/div (A/div) and offset that make the trace fill most of the screen without running off it — measurements on a clipped trace are wrong. Put the zero level on a graticule line and note it on your captures. Lab handouts give suggested starting values.
- Coupling: keep DC coupling for averages. To look at ripple riding on a large average, switch that channel to AC coupling and reduce V/div (or A/div); switch back to DC before reading an average.
- Steady-state waveforms: edge trigger on a switching waveform such as the inductor current, level near its average, mode Auto.
- Slow transients (for example a reference or load step): edge trigger on the signal that changes, level just past its starting value, mode Normal. Turn the horizontal position knob so the trigger point sits about two divisions from the left edge, choose a time/div that fits the whole response, press [Single], then make the change.

Measurements

- **Use the scope's measurements rather than reading values off the graticule.** [Meas] → choose the Source channel and the Type, e.g. Average, DC RMS, Peak-Peak, Maximum, Minimum, Frequency, Duty Cycle. The N Cycles variants find a cycle from the source's own edges; on a PWM or inverter waveform these are the switching edges, not the fundamental. To measure over whole fundamental periods, use the Full Screen variant with a whole number of periods on screen (5 ms/div shows exactly three 60 Hz periods).
- Cursors for time intervals and levels: [Cursors] → place X1 and X2 on two edges and read ΔX (for example a conduction interval or a settling time); Y1 and Y2 read levels.

FFT (spectrum)

- Setup: [Math] → Function f(t) → Operator FFT, and Source 1 = the channel to analyse. More FFT → Window → Flat Top, which gives the most accurate peak

amplitudes (Hanning is better for locating a frequency precisely). Flat-top peaks are several bins wide; this is expected, so read the top of the peak.

- Units: each peak is the rms value of that sinusoidal component, in dB. 0 dB is 1 V rms (dBV) on a volts channel, or 1 A rms on a channel whose units are Amps. A peak at L dB is 10^(L/20) rms: −20 dB = 0.1 A, −40 dB = 10 mA. The FFT uses the channel's probe setting, so set the current channel as under Probes first; otherwise every dB value is off by the setting error, e.g. 20 dB low if the channel is left at Volts, 1:1.
- Time/div sets the resolution. The FFT uses the time record on screen, and its resolution (the bin spacing) is about one over the record length: 20 ms/div gives a 200 ms record and about 5 Hz. The actual value is shown above the softkeys. Keep at least ten fundamental periods on screen so the harmonics of 60 Hz are separate peaks.
- Source scale: set the source channel so its time-domain trace nearly fills the screen without clipping, even if you turn the channel's display off. A clipped source corrupts every line of the spectrum; a trace that is too small buries small components in the noise floor.
- Span and Center select the band shown; Scale (dB/div) and Offset place the peaks on screen. Start from 10 dB/div and a span that covers every component you expect. Once you set Scale or Offset by hand, Span and Center stop following time/div; More FFT → Auto Setup restores the full spectrum.
- Reading a peak: press [Stop] (or [Single]) to freeze the record, then [Cursors] → Source Math f(t). Put X1 on the peak to read its frequency and Y1 on its top to read its level. Narrow the span around the peak for finer cursor placement. Check that the frequency is one you expect before recording the value.
- **Pitfalls.** Ignore the DC bin: the scope does not correct it for the channel offset. Components above half the sample rate fold back into the displayed band (aliasing): if a peak appears where none is expected, change time/div by one step. A real component stays at the same frequency; an alias moves. Peaks within a few dB of the noise floor are not reliable; report them as below that level. Changing the window changes peak heights, so use Flat Top for every value you report.

5. Fluke 39 — power harmonics meter

- The Fluke 39 measures single-phase volts, amps and watts at a 6–65 Hz fundamental and analyses harmonics from DC to the 31st (1.86 kHz at 60 Hz). Its input bandwidth is about 2.1 kHz.
- Connections: voltage leads into V (red) and COM (black), 5 V to 600 V rms. Current input: the BNC CURRENT PROBE jack, fed from the Fluke i400s AC current clamp. The 1146B goes to the oscilloscope only: clamp the i400s around the same wire, with the arrow on its jaw pointing in the direction of positive current.
- **The meter assumes a 1 mV/A clamp.** The i400s has two ranges, chosen with its switch: 40 A (10 mV/A, for 0.5 A to 40 A) and 400 A (1 mV/A, for 5 A to 400 A). Use

the range the lab handout gives. On the 40 A range every current reading (A) and power reading (W, VA, VAR) is 10 times the true value: divide by 10. On the 400 A range the readings need no correction. Volts, PF, DPF, THD, percentages and phase angles are unaffected. The i400s measures AC only and needs no battery or zeroing. The current input needs at least 1 mV rms and accepts up to 2 V peak; beyond that it shows OL-A.

- **Check the setup screen every session:** the settings are stored in the meter and the previous user may have changed them. Hold [HOLD/ENTER] while pressing the power key. ◄ ► highlights a choice and [HOLD/ENTER] accepts it and moves to the next line; accepting the last line stores the setup. Set THD to %F (THD relative to the fundamental; the factory setting %RMS gives THD relative to the total rms), CLAMP to OTHER (the 80I-500S setting corrects for a different Fluke clamp; OTHER gives the flat response the i400s needs) and Ø REF to VOLT.
- Keys: the V/A/W key selects the measurement (volts, amps or watts); the screen-mode key beside it cycles waveform, harmonics and text screens. The text screen lists rms, peak, DC and THD for V and A, and W, VA, VAR, PF and DPF for watts. [HOLD] freezes the screen for reading or a photo. [SMOOTH] averages over 2, 5, 10 or 20 s; use 20 s for harmonic readings.
- Harmonics screen: ◄ ► moves the cursor along the bars (DC and 1–15 on the first page, 16–31 on the second). The top line shows the selected harmonic's value, its percentage of the fundamental (with THD set to %F) and its frequency. [RANGE] zooms the bar scale (100, 50, 20, 10 %); it changes the display only. The harmonics screen shows an overload message if the reference input (the voltage leads, with Ø REF = VOLT) has no usable signal; the voltage input needs at least 5 V rms.
- PF is W/VA; DPF is cos φ of the fundamental alone. Accuracy: harmonic currents ±(3 % of reading + 3 digits) and THD ±(3 % of reading + 2 percentage points) with 20 s smoothing, so a THD of a few percent, or a harmonic of a few tens of mA, is only roughly known.

6. Acuvim-L — transformer panel power meter

- The Acuvim-L (model Acuvim-CL) on the station transformer panel measures three-phase volts, amps, power, power factor and harmonics to the 31st. Each phase current comes from a 30:5 A window CT (ITI 2SFT-300) into the meter's 5 A input. Voltages and currents are true rms (fundamental plus harmonics).

Display modes

- Hold [H] and [V/A] together for about one second and release. The mode list (Meter, Max/Min, Demand, Harmonic, Setting, Digital I/O) appears with a flashing cursor; [P] and [E] move the cursor and [V/A] enters the selected mode. Digital I/O only shows I/O-module data.

- **Do not enter Setting.** Its pages are password protected and hold the CT and wiring settings; a wrong value there makes every reading wrong. Report any reading that looks wrong to the TA.

Meter mode

- The first page is a summary: average voltage, total power, average current, total power factor and total energy.
- [V/A] steps through the voltage and current pages, [P] through power, power factor and frequency (P, Q and S per phase, then the system totals), [H] through phase angles and unbalance, and [E] through energy. Rows 1, 2 and 3 are phases A, B and C; avg is the average. With line-to-line wiring settings (3LL, 2LL) only line voltages and system totals are shown.
- The inductor or capacitor icon shows whether the load is inductive or capacitive at that moment.
- **Power factor.** Each phase PF is that phase's P divided by its line-to-neutral voltage times its rms current, and the total PF is the total P divided by the sum of the phase volt-amperes. For a balanced three-phase load the total PF is the true PF and can be read directly. For a single-phase load connected line-to-line (for example A–B), each phase PF is shifted by about 30° and the total reads about 0.87 ($\sqrt{3}/2$) of the true value: calculate PF = P / (V_AB · I_A) from the meter's total P instead, as the lab handout asks.

Harmonic mode

- The first pages are power-quality data: [H] steps through voltage THD, odd and even harmonic distortion and crest factor, then current THD, odd and even harmonic distortion and K factor. The page is identified by the U or I symbol and the THD label; the unit is %.
- [V/A] switches to individual harmonics. [P] and [E] step the harmonic order up and down (the display shows, for example, I 03 for the 3rd current harmonic), and [V/A] switches between voltage and current. Each value is a percentage of the fundamental, not amps or volts.
- THD is relative to the fundamental, as in class. To convert a harmonic to amps: I_1 = I(rms) / $\sqrt{(1 + \text{THD}^2)}$, with THD as a fraction, then multiply I_1 by the harmonic's percentage / 100.

Accuracy

- The manual gives ±0.5 % for V, I, P and PF and ±1 % for harmonics, for the meter alone. The CTs add their own error, which is largest at low current: at a few amps each CT carries less than 10 % of its 30 A rating, where its accuracy is not specified and it tends to read low. When an rms current matters, check it with the 1146B on the oscilloscope.

Settings check (TA or instructor)

- Setting → password (default 0000) → System. [P] and [E] page through the settings; [V/A] edits a value, [H] moves the flashing digit and [P]/[E] change it, and [V/A] stores it.
- Required values: S04 voltage wiring to match the station wiring (3LN for a four-wire wye); S05 current wiring 3CT; S08 CT1 = 30 (the CT primary in amps; if the conductor passes through the CT window N times, use 30/N); S09 CT2 = 5; S31 Basic Parameter Mode PRE (SEC shows the CT secondary current instead); S35 Energy Type Full.
- Symptoms: currents about 1/6 of the true value mean CT1 is still at its default of 5 or S31 is set to SEC; five times too high means CT2 = 1. A phase with negative power has a reversed CT: turn the CT around or set that phase's direction (S21–S23) to Negative. S55–S58 map each current input to its phase if phases are swapped.

7. Resources

Resource	Where
E3631A user's guide	https://www.keysight.com (search E3631A User's Guide)
BK Precision 1665/1666/1667 manual	https://bkpmedia.s3.amazonaws.com/downloads/manuals/en-us/1667_manual.pdf
ITECH IT-M3600 series user manual	https://www.itechate.com (IT-M3600 User Manual)
InfiniiVision 2000 X-Series user's guide	https://www.keysight.com/us/en/assets/9018-03426/user-manuals/9018-03426.pdf
Fluke 39/41B users manual	https://www.fluke.com (search 39/41B Users Manual)
Acuvim-L user manual	https://www.accuenergy.com/wp-content/uploads/Acuvim-L-Multifunction-Power-and-Energy-Meter-User-Manual-v4.pdf
Station equipment list	Lab Resources (on D2L)